

6 $p^3 \times p^x = p^9$

(a) Find the value of x .

$x =$
(1)

$(7^2)^y = 7^{10}$

(b) Find the value of y .

$y =$
(1)

$100^a \times 1000^b$ can be written in the form 10^w

(c) Show that $w = 2a + 3b$

(2)